CSE 123A - Demo Script
Water Level Monitoring System

Group 8

Part 1: Introduction

Presenter: Pieter
Estimated Time: 1:30 minutes
Action: Hold up or gesture toward the water pitcher.

Script:

> "Has this ever happened to you? You come home after a long day, you're thirsty, and you go to grab the water filter pitcher—only to find it completely empty. Nobody refilled it. This is a common problem in shared living spaces—dorms, family homes, offices. Everyone uses the filtered water, but nobody knows when it's running low until it's too late. Our project solves this. We want a way to know if the container is out of water when we need water. Our goal is that if the container is empty, users get informed. We built a smart water level monitoring system that sits underneath your water filter pitcher, continuously tracking how much water is left, and sends a notification to everyone in the household when it's time to refill."

Part 2: Overview

Presenter: Reid
Estimated Time: 1 minute
Action: Point at the various system components as they are mentioned.

Script:

> "Our design has three main components: the smart base, the web server, and the mobile app.
>
> The smart base sits beneath the water pitcher and measures the weight of the pitcher, which can be used later to determine how much water is left. This weight data is sent to our web server over WiFi using HTTP packets, where it's stored in a database and processed.

The mobile app is connected to the web server and is able to access this weight data. When the water level drops below a certain threshold, the mobile app will send push notifications to the users. The app also allows users to view the current water level and configure various settings.

[Switch to prototype diagram.] Today, we will be demonstrating the prototype of our design. We are simulating the connection between the smart base and the web server through two laptops: one connected to the microcontroller to read the sensor data, and another sending simulated weight data to the web server. Our mobile app is currently in the form of a webapp on a laptop, but still has the core functionality that the mobile app would have."

Part 3: Hardware & Live Demo

Presenters: Dev & Logan
Estimated Time: TODO
Action: Dev places the pitcher on the platform with the terminal open on the connected laptop. Logan picks up or points to each hardware component as he explains it.

Script:

Dev:

"Let me set up the live demo first. I'm going to place this empty water pitcher on the sensor base. Over here on the laptop we have the terminal open, where you can see the raw weight readings coming in from the microcontroller via USB. The load cell is now measuring the weight of the pitcher.

While this sits on base, Logan is going to walk you through the hardware that makes this work."

Logan:

So let me break down what's going on inside this base. The system has three main components:

First, the load cell. This is a strain-gauge weight sensor. When weight is applied, it bends slightly, and that bending changes its electrical resistance. The change is very tiny—so we use the **HX711 amplifier board** to boost the signal into something our microcontroller can read. The HX711 gives us 24-bit resolution, which means we can detect very small changes in weight.

Second, the microcontroller. This is the brain of our system. It reads the weight data from the HX711 and outputs the readings to the terminal you see on the laptop.

Third, the mounting base. We designed and 3D-printed two circular plates that sandwich the load cell. The pitcher sits on top. We designed these with M4 screw holes so everything stays secure and aligned."

Dev:

> "Now that you know how the hardware works, let me show you how it reads in action. So right now the terminal is showing the current weight value with the pitcher on the base.
>
> So now let me pour some water in to simulate a user putting water in the pitcher.
>
> *[Pour water into the pitcher.]*
>
> Now if you watch the terminal, you can see the weight values increasing as we pour water into the pitcher. The microcontroller is reading the load cell continuously, so the changes show up right away." Now let me pour some water out to simulate someone using the pitcher throughout the day.
>
> Now let me pour some water out to simulate a user using some of the water from the pitcher. *[Pour water out of the pitcher into a cup.]*
>
> And just like before, you can see the values in terminal drop as the water after we put the pitcher back on the base.

Part 5: Software Demo

Presenter: Shriyan
Estimated Time: TODO
Action: Open laptop to the web server to demonstrate navigating the page

Script:

> "The initial prototype uses a web dashboard to display important device information. This dashboard is also where the initial configuration of the product happens. When a user first starts up this device, they need to calibrate it with empty and full weights.
>
> *[Highlight calibration buttons]* Both an empty and a full value are required for the product to work as intended. In the startup sequence the user will be prompted for an empty pitcher of water to be placed on the base. After pressing the calibration button they will be asked to fill it up and return it to the base to calibrate the full value.
>
> Full and empty values are adjustable and configurable at any time, so if a user buys a new filter that holds a larger quantity of water, they can simply change the calibration values and continue using the device as intended.
>
> When the values are calibrated, the system is able to use the data sent by the microcontroller to adjust the visible water level."

Part 6: Notification Demo

Presenter: Sam
Estimated Time: TODO
Action: Keep the dashboard visible and reset the water level to above the low-water threshold 20%.

Script:

> "Now we will demonstrate the push notification feature and how we can subscribe to notifications for when the water level is running low.
>
> *[Point to the current percentage on the dashboard.]* As you can see here on the dashboard, right now, the water level is at [read water level value]. Our low-water threshold is currently set to 20% and as you can see, the pitcher level is above 20% right now, so no notifications will be sent. Once the level drops below that threshold, the web app will trigger a push notification to subscribed devices. But to see this happen, we have to first subscribe to notifications.
>
> *[Point to "Enable Notifications" button on the dashboard.]* Once you click on this button here that says "Enable Notifications", it will prompt you to allow notifications for our web app. Once you press allow on the prompt, your device will be subscribed and ready to receive notifications. Now lets try clicking on the button and allowing notifications.
>
> *[Click on "Enable Notifications" button and allow]* Ok, now that we have enabled notifications, you can see on the dashboard under the Water Pitcher Status that it says "Notifications are enabled". Now, we can test it out by first measuring the pitcher when it is near empty. [Water Refiller Person] will now pour out water from the pitcher until it is almost empty and put it back on the base.
>
> *[Pour out water from the pitcher]* And since our base is currently not integrated with the webapp, we will simulate this change by manually sending a low weight value to the web server to trigger the notification. Once the low weight value is sent, you should see the dashboard update to show the new water level and you will also be able to see the push notification appear on the top right corner of the screen, so please watch closely. We will now send the low weight value.
>
> *[Simulate water level to less than 20%]* As you can see on the dashboard, it now reads [read value below 20%]. You should've also seen the push notification pop up on the top right corner of the screen. If you were paying attention, you would've seen that the message payload in the notification showed that the filter is running low and needs to be refilled. It also showed the current water level percentage. Now [Water Refiller Person] will refill the pitcher to a percentage that is above the low-water threshold.
>
> *[Refill the pitcher to above the low-water threshold and put on base]* After the pitcher is refilled and the percentage rises, the system returns to normal and waits for the next low-water event."

Part 7: Questions

Presenter: Pieter (prof suggested same as intro person)
Estimated Time: 5 minutes
Action: Leave all components visible on the center of the demo table, ready to be moved or pointed at for any questions regarding hardware.